

$$① \quad b_{yx} b_{xy} = r^2 \Rightarrow r = \pm \sqrt{b_{yx} b_{xy}}$$

~~$$② \quad b_{yx} = \frac{1}{b_{xy}} \quad r^2 \leq 1$$~~

~~$$b_{yx} b_{xy} \leq 1$$~~

~~$$\Rightarrow b_{yx} \leq \frac{1}{b_{xy}}$$~~

1. If b_{yx} is greater (less) than one and vice versa.

Pt. but $b_{yx} > 1 \Rightarrow \frac{1}{b_{yx}} < 1$

now $r^2 \leq 1 \Rightarrow b_{yx} b_{xy} \leq 1 \Rightarrow \underline{\underline{b_{xy} \leq \frac{1}{b_{yx}} < 1}}$

Remarks

Appointments

3/ Regression coefficient is affected.
is invariant under origin change
but affected by scale change.

$$\text{Let } u_i = \frac{x_i - \bar{x}}{d}, \quad v_i = \frac{y_i - \bar{y}}{d'}$$

$$\therefore \text{Cov}(u, v) = dd' \text{Cov}(u, v)$$

$$\text{and } \sigma_u^2 = d^2 \sigma_u^2, \quad \sigma_v^2 = d'^2 \sigma_v^2$$

Phones to make

$$\therefore b_{yx} = \frac{\text{Cov}(x, y)}{\sigma_x^2} = \frac{dd' \text{Cov}(u, v)}{d^2 \sigma_u^2}$$

$$= \frac{d'}{d} b_{uv}$$

$$b_{yx} = r_{xy} \frac{\sigma_y}{\sigma_x} = r_{uv} \frac{d'}{d}$$

Remarks

$$\text{Similarly } b_{xy} = \frac{d}{d'} b_{uv}$$